Department of Computer Science
Faculty of Computing

Project 1

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1.0 INTRODUCTION

Obesity is one of the main health issues faced by many people today. It is closely linked to other serious health problems, especially diabetes, hypertension and heart disease. These three conditions are often called the triad of diseases because they are strongly connected and commonly occur together. If not controlled early, they can lead to long term health complications and even shorten a person’s lifespan.

In this project, we use a dataset that includes information about people’s lifestyle habits and physical characteristics, such as age, gender, weight, height, daily activity and eating patterns. The main aim is to build a model that can predict a person’s obesity level, which is labeled as NObeyesdad, using supervised learning techniques. This means we train a model using data that already includes the correct obesity level, so the model can learn and then make predictions on new data.

We follow a structured data mining process that includes understanding the dataset, preparing the data, building the prediction model, and checking how well the models work. The software used in this project is RapidMiner, and we evaluate the model’s performance by using four key metrics which are accuracy, precision, recall and F1-score. By completing this project, we hope to show how data mining can help detect obesity early and support better health decisions for preventing the triad of diseases.

2.0 BUSINESS AND DATA UNDERSTANDING

This section outlines the primary objective of the project and explains how the dataset is utilized to support this goal. It is helpful to understand the project's purpose and the importance of the data before building any prediction model.

2.1 PROBLEM FORMULATION

Obesity, along with diabetes and hypertension, forms a dangerous health triad that affects millions of people worldwide. These conditions are often caused by poor lifestyle habits, such as unhealthy eating, lack of exercise, and other behavioral factors. With the help of data mining techniques, we can analyze people's habits and physical features to detect early signs of obesity.

The main problem addressed in this project is how to **predict a person's level of obesity** using data about their lifestyle and body measurements. By building a supervised learning model, we aim to classify each individual into one of several obesity categories, such as Underweight, Normal Weight, Overweight, Obesity Type I, Obesity Type II, and Obesity Type III. This prediction can help in early health monitoring and promote better decisions for preventing long term health issues.

2.2 DATASET OVERVIEW AND ITS IMPORTANCE

The dataset used in this project is "ObesityDataSet_raw_and_data_sinthethic.csv" is retrieved from UCI Machine Learning Repository platform with the keyword "Obesity" which contains both real and synthetic data about individuals. It includes 17 input features, such as age, gender, height, weight, family history, eating habits, physical activity level, smoking, and technology usage. The target attribute is NObeyesdad, which represents the obesity category of each person.

This dataset is important because it provides a complete view of various factors that may lead to obesity. It allows us to explore how behavior and body metrics are linked to weight status. Since the data includes many relevant attributes, it is suitable for training machine learning models that can help in predicting obesity levels. This can be very useful for health professionals, educators and the general public to understand risks and take action before health conditions become more serious.

3.0 DATA PREPARATION AND PROCESSING

For the dataset used in this project, which is "ObesityDataSet_raw_and_data_synthethic.csv", we went through several preprocessing steps to process raw data to make it suitable for analysis. For this data preparation and processing, we clean the data to ensure the data is of quality without any incomplete, noisy, or inconsistent data.

3.1 OVERVIEW AND ITS IMPORTANCE

Data preparation and processing are crucial steps in the data mining workflow. These steps ensure that the raw data is transformed into a format suitable for analysis and model building. In this project, the data preparation phase involved cleaning the dataset, handling missing values, normalizing numerical features, and converting categorical attributes into numerical format. This phase is vital because the quality of the input data directly impacts the performance and reliability of the machine learning model. By addressing data inconsistencies and ensuring data integrity, we lay a solid foundation for developing an effective predictive model for obesity levels.

3.2 DATA CLEANING AND PREPROCESSING STEPS FOR SUPERVISED LEARNING

Data Cleaning

The initial step in data preparation was data cleaning. We examined the dataset for missing values, duplicates, and inconsistencies. Fortunately, the dataset provided did not contain missing values or duplicates. However, we did identify some inconsistencies in the data types of certain attributes. For instance, some numerical attributes were initially stored as strings. We corrected these inconsistencies by converting the data types to appropriate numerical formats.

Handling Categorical Data

The dataset contained several categorical attributes such as gender and family history. These categorical variables needed to be converted into a numerical format to be compatible with the k-NN algorithm used in this project. We employed one-hot encoding to transform these categorical features into binary vectors, ensuring that the machine learning algorithm could process them effectively.

Normalization

Normalization is a critical preprocessing step, especially for distance-based algorithms like k-NN. We normalized all numerical features to a common scale, typically between 0 and 1. This step ensures that features with larger value ranges do not disproportionately influence the distance calculations, thereby improving the model's performance.

Feature Selection

We carefully selected the most relevant attributes for model training. This involved removing features that were deemed irrelevant or redundant for predicting obesity levels. By focusing on pertinent features such as age, weight, height, physical activity level, and eating habits, we aimed to enhance the model's efficiency and accuracy.

Data Splitting

The dataset was split into training and testing sets using the Split Data operator in RapidMiner. We allocated 70% of the data for training the model and reserved 30% for testing its performance. This split allows us to evaluate how well the model generalizes to unseen data.

3.3 RESULTS OF DATA PREPARATION TASKS

After completing the data preparation tasks, we obtained a cleaned and preprocessed dataset ready for model training. The dataset was free from inconsistencies and missing values, with all features appropriately formatted and scaled. The feature selection process resulted in a streamlined dataset containing only the most relevant attributes for predicting obesity levels. The split data was then used to train and evaluate the k-NN model, yielding promising results as detailed in the subsequent sections of this report.

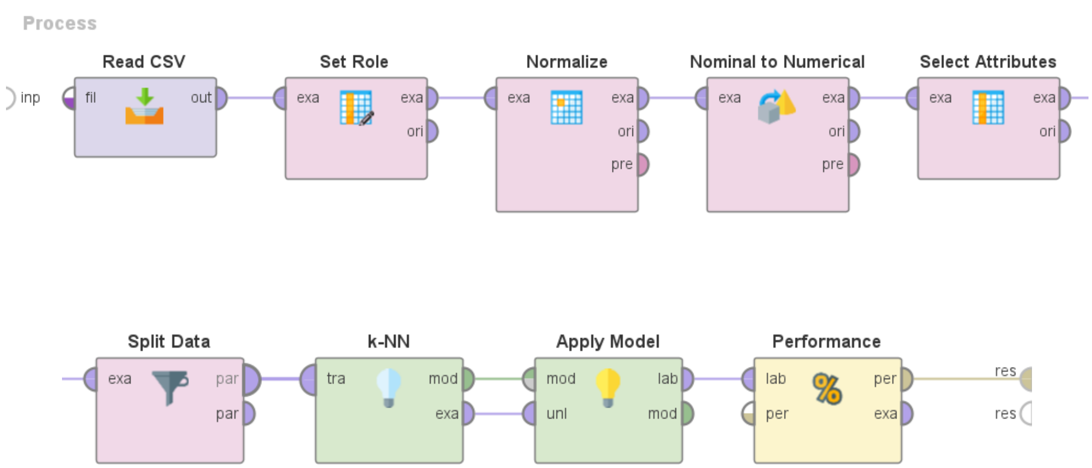
4.0 MODEL DEVELOPMENT (SUPERVISED LEARNING)

4.1 OVERVIEW OF SUPERVISED LEARNING METHOD

Supervised learning is a machine learning approach where the model is trained using labeled data. This means the algorithm adapts and learns the relationship between input features and a known output which is label or class, so it can later predict the label for new unseen data. In this project, we used K-Nearest Neighbours (k-NN) as our supervised learning algorithm to predict the obesity level of individuals. The k-NN algorithm is a simple but effective classification algorithm. It works by finding the closest 'k' training examples in the feature space and assigning the most common class among them to the test example. It does not assume any specific distribution in the data.

4.2 APPLIED ALGORITHM AND PROCESS WORKFLOW

We built the supervised learning model using RapidMiner. Below are the main steps in the process:



1. Read CSV: The obesity dataset was imported into RapidMiner.
2. Set Role: The target label Nobeyesdad, was set as the label.
3. Normalize: All numerical features were normalized to ensure fair distance calculation in k-NN.
4. Nominal to Numerical: Categorical features were converted into numerical format to work with the k-NN algorithm.

5. Select Attributes: Only relevant attributes were selected for model training.
6. Split Data: The data was split into 70% for training and 30% for testing using the Split Data operator.
7. k-NN Model: The model was trained on the training set and tested on the test set.
8. Apply Model: The trained model was applied to the test set.
9. Performance Evaluation: The model's performance was evaluated using a confusion matrix that gave accuracy, precision, and recall.

4.3 MODEL RESULTS AND EVALUATION

The model achieved the following performance:

1. Accuracy: 83.76%

accuracy: 83.76%

	true Normal_Weight	true Overweight_Level_I	true Overweight_Level_II	true Obesity_Type_I	true Insufficient_Weight	true Obesity_Type_II	true Obesity_Type_III	class precision
pred. Normal_Weight	103	3	3	0	4	1	1	89.57%
pred. Overweight_Level_I	18	174	3	9	1	1	0	84.47%
pred. Overweight_Level_II	20	8	151	8	3	2	1	78.24%
pred. Obesity_Type_I	24	13	11	209	4	6	0	78.28%
pred. Insufficient_Weight	18	0	5	0	174	1	0	87.88%
pred. Obesity_Type_II	13	7	18	12	1	208	0	80.31%
pred. Obesity_Type_III	7	4	2	2	6	0	219	91.25%
class recall	50.74%	83.25%	78.24%	87.08%	90.16%	94.98%	99.10%	

2. Weighted Mean Recall: 83.36%

weighted_mean_recall: 83.36%, weights: 1, 1, 1, 1, 1, 1, 1

	true Normal_Weight	true Overweight_Level_I	true Overweight_Level_II	true Obesity_Type_I	true Insufficient_Weight	true Obesity_Type_II	true Obesity_Type_III	class precision
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class recall	50.74%	83.25%	78.24%	87.08%	90.16%	94.98%	99.10%	

3. Weighted Mean Precision: 84.28%

weighted_mean_precision: 84.28%, weights: 1, 1, 1, 1, 1, 1, 1

	true Normal_Weight	true Overweight_Level_I	true Overweight_Level_II	true Obesity_Type_I	true Insufficient_Weight	true Obesity_Type_II	true Obesity_Type_III	class precision
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5.0 MODEL PERFORMANCE & EVALUATION (ACCURACY, PRECISION, RECALL, F1)

5.1 APPLIED EVALUATION METRICS

In this project, the performance of the K-Nearest Neighbors (K-NN) model was evaluated by using several evaluation metrics. Since the classification task involves multiple classes, it is crucial to use a collection of metrics to find the model's predictive ability. Furthermore, we are using a confusion matrix to provide a detailed of the model's classification results for each class.

- **Accuracy** : measure the accuracy of the model to predict the label correctly by calculating the ratio of the correctly predicted case to the total case
- **Precision** : measure the correctness to the predicted case of each class out of all cases predicted as the class. **Weighted mean precision** is used to summarize the performance of all classes.
- **Recall** : measure how much the actual case of the class was correctly identified by the model. **Weighted mean Recall** is used to summarize the performance of all classes.
- **F1-Score** : The harmonic means that it provides a balance between precision and recall. This can help with the imbalance of the classes.

5.2 PERFORMANCE ANALYSIS OF THE MODEL

Metric	Value
Accuracy	83.76%
Weighted Mean Recall	83.36%
Weighted Mean Precision	84.28%

5.2.1 ACCURACY ANALYSIS

The model has an overall accuracy of 83.76% that indicates that out of 100 cases the model correctly predicts 84 cases. This shows that the model learned the relationship between input features and the label well.

5.2.2 PRECISION AND RECALL ANALYSIS

The weighted mean precision of 84.28% shows that when the model predicts a certain obesity level. The correctness of the prediction is 84% out of all the cases. Furthermore, The weighted mean recall of 83.36% indicates that out of all actual cases of a particular obesity level, the model can correctly identify 83% of them. This balance between the precision and recall shows that the model performs consistently well in different classes of obesity by minimizing both false positives and false negatives.

5.2.3 F1-SCORE

The weighted F1-score occurs when combining both precision and recall into a single metric, providing a balanced view of the model's performance. This is how we calculate the F1-Score:

formula F1-Score :

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Precision = weighted mean Precision = 0.8428

Recall = Weighted mean Recall = 0.8336

$$F1 = 2 \times \left(\frac{0.8428 \times 0.8336}{0.8428 + 0.8336} \right) = 0.8381 = 83.81\%$$

The F1-score of 83.81% indicates that the model maintains a good balance between precision and recall across the classes and making it effective for multi-class classification.

5.3 OVERALL MODEL PERFORMANCE SUMMARY

The K-Nearest Neighbors (KNN) model successfully shows a strong ability to predict the obesity levels based on the provided dataset. With accuracy, precision, recall and F1–score are in the range of 83-84%, it indicates that the model shows a good classification ability without being biased toward any classes. This Performance indicates that this model can be use in real-world applications for early obesity detection , personalized health recommendation and public health monitoring

6.0 CONCLUSION

In this project, a supervised learning model was developed to predict the obesity levels based on the lifestyle habits and physical characteristics of the individual. We are using a dataset obtained from the UCI Machine Learning, which includes attributes such as age, weight , height, eating habits and physical activity. The goal of the project is to classify the individuals into six categories which are Underweight, Normal Weight, Overweight, Obesity Type I, Obesity Type II and Obesity Type III.

The Dataset was preprocessed through data cleaning, normalization and conversion of categorical attributes into numerical form. The K-Nearest Neighbors (K-NN) algorithm was applied using a RapidMine where 70% of the data was used for training and 30% for testing. The model performance is evaluated using a confusion matrix where we use it to find the accuracy, precision, recall and F1-score.

The K-NN model has an accuracy of 83.76%, precision (weighted mean precision) of 84.28%, a recall (Weighted mean Recall) of 83.36%, and F1-score of 83.81%. These results indicate that the model performs reliably across multiple classes and can be used for early detection and monitoring of obesity related health risks.